

Business Requirement Document

(BRD)

Project Name: CrowdFlow

Group Number: Group 7

Date: June 10th, 2026

Version: 1.1

Team Members

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Document Control

Version History

Version	Date	Author	Changes
1.0	June 10th, 2026		Initial draft
1.1	June 16th, 2026 - TBA		First Revision

1. Executive Summary

Provide a concise overview of the project for decision-makers who may not read the full document. Keep it to one page maximum. Tip: Write this section last, after all other sections are complete.

To eliminate the structural and financial losses caused by automated scalping bots which account for up to 73% of high-demand ticketing traffic and trigger catastrophic server instability, this project introduces a centralized and an edge-to-core secured ticketing platform. The solution deploys behavioral bot prevention, a virtual waiting room queue, and local payment gateway integrations to mitigate immense infrastructure strain while ensuring stable database operations for over 1,000 concurrent users. To completely neutralize the secondary resale market and guarantee fair distribution, the system mandates that ticket purchases be linked to government identification (KTP/Passport) and validated via cryptographically secure, single-use QR codes at venue gates, while offering a platform-managed, capped-price customer-to-customer resale marketplace. Governed by a rigid Role-Based Access Control (RBAC) model across a five-phase development lifecycle concluding on July 15th, 2026, this architecture successfully safeguards platform operations, protects consumers from fraud, and maximizes ecosystem profitability.

1.1 Business Problem / Opportunity

Briefly describe the business problem or opportunity that this project addresses.

The online ticketing industry is facing a critical operational crisis driven by sophisticated, industrial-scale automated bots and scalpers. Malicious automated scripts account for an average of 40% of all ticketing platform traffic, systematically bypassing transaction limits and mass-creating fake accounts to harvest premium inventory within seconds of a public release. This ongoing digital siege forces primary vendors to bear extreme server infrastructure costs to handle artificial traffic peaks, while transferring the economic surplus of live events away from artists and platforms into an unregulated secondary resale market.

The vulnerability of modern infrastructure was starkly illustrated during Ticketmaster's 2022 Taylor Swift Eras Tour presale. Scalping bots bombarded application endpoints with an unprecedented influx of system requests, peaking at four times the platform's historical traffic high. This automated assault triggered severe server instability, corrupted checkout cart database sessions, and locked out millions of verified fans, ultimately forcing the cancellation of the public sale. Modern cybercriminals continuously engineer advanced bypass scripts capable of mimicking human biometrics, with recent reports estimating that automated attacks can constitute up to 73% of total traffic during high-demand on-sale windows.

For online vendors, these automated disruptions inflict deep financial and structural losses. When bots hoard ticket inventory into temporary checkout carts, they create severe database deadlocks and artificial stockouts. If the secondary market fails to absorb this scalped inventory, vendors are left with empty venue seats and a steep decline in critical day-of-event revenue streams like merchandise, concessions, and parking. Furthermore, when genuine consumers are forced onto third-party resale sites where tickets are highly inflated or fraudulent, the primary platform bears the brunt of the fallout that results in severe brand degradation, user churn, and costly regulatory scrutiny.

1.2 Proposed Solution

State the proposed solution in 2 to 3 sentences.

To mitigate the revenue loss, infrastructure strain, and trust erosion caused by automated scripts and fragmented operations, this platform introduces a centralized ticketing ecosystem engineered around edge-to-core security, dynamic scale management, and operational integrity. The solution deploys behavioral bot prevention, a virtual waiting room queue, and dynamic QR codes to neutralize scalper scripts, prevent server crashes, and eliminate ticket duplication during high-velocity on-sales. Governed by a strict multi-tiered RBAC model and a single back-office operational space, this unified architecture streamlines onboarding, integrates secure payment gateways, and validates events to protect consumers from fraud while maximizing platform profitability.

1.3 Key Stakeholders

Mention key stakeholders involved in the project.

Stakeholder	Responsibility
Users	Purchase and manage event tickets.
Event Organizers	Create and manage events and ticket sales
Vendors / Venue Managers	Manage venues, seating arrangements, and capacity
Event Auditors (Admin)	Verify, review, and approve events.
Super Admin	Manage platform operations and user access

1.4 Project Timeline

Mention the overall project timeline or key milestones.

Phase	Description	Start Date	End Date	Milestone
Discovery	Gather requirements, Completing BRD Document, Finding	9 June 2026	19 June 2026	Project Scope Realised

	reference for Website structure			
Design	Creating UI/UX wireframes, user and admin dashboard sketch and database structure	12 June 2026	21 June 2026	w
Development	Implement the wireframes and sketch from design phases into a working website	20 June 2026	12 July 2026	
Testing (UAT)	Giving limited access of the work-in-progress website to the user for feedback and criticism	1 July 2026	12 July 2026	
Deployment	Deployment of website from github	10 July 2026	15 July 2026	

2. Project Scope

Define the boundaries of the project to prevent scope creep and set clear expectations.

2.1 In-Scope

List the features and functionality that will be delivered as part of this project.

1. Secure account creation, login, and session management, including collection of verified user information such as email address, legal name, identification documents (KTP or Passport), and bank account details for eligible transactions.
2. Forget password mechanism: A self-service workflow allowing users to securely reset their password via email validation
3. CAPTCHA-based bot prevention: Automated challenge-response checkpoints at checkout to filter out malicious script and bots
4. Event verification and auditing: A workflow requiring organizers to submit legal letters for auditors to review and approve before an event goes public
5. Venue management and seating arrangement: Tools or libraries to digitize venue layouts, map sections/rows, and support both reserved seating and standing room setups
6. Unique QR Code ticket generation and validation: Cryptographically secure, single-use QR codes generated per ticket and validated via a scanning interface at the venue gate.
7. Online payment gateway integration: Integration with a local provider to process credit cards, bank transfers, and e-wallets smoothly.
8. Queue and waitlist system for high-demand events: A virtual waiting room that throttles high traffic during peak demand to protect database infrastructure from crashing
9. Role-Based Access Control (RBAC): Strict data security rules ensuring users only access features permitted by their specific tier (User, Vendor, Organizer, Auditor, Super Admin).
10. User dashboard and ticket management: A clean storefront interface where customers can view purchase history, manage active tickets, and quickly pull up their QR codes.
11. Centralized administration dashboard: The core back-office control panel allowing platform admins to manage users, monitor revenue, and audit system configurations
12. Venue Capacity Management: Guardrails tied to the database that strictly enforce maximum safety caps based on real-world venue dimensions to prevent overselling.
13. Identity Registration and Event Check-In Verification: A ticket holder registration process that requires users to assign KTP or Passport information to each ticket holder. Identity verification will be performed by event staff during event check-in.
14. Ticket Purchase Limitation: Configurable purchase limits that allow Vendors or Event Organizers to define the maximum number of tickets that can be purchased per account for each event to reduce ticket hoarding and scalping activities.
15. Official Ticket Resale Marketplace: An integrated customer-to-customer exchange that allows verified users to securely resell previously purchased tickets through the platform at prices that do not exceed the official ticket price.

2.2 Out-of-Scope

Explicitly state what is NOT included in this project.

1. Physical ticket printing services. No infrastructure for printing thermal paper, tickets, mailing wristband, or managing box-office hardware
2. Transportation booking services. No integration with rideshares, flights, or trains to help users get to the venue.
3. Hotel or accommodation booking. No lodging packages, hotel partnerships, or hospitality upselling features at checkout.
4. Event marketing and advertising services. No built-in ad networks, email marketing campaign tools, or public promotional banners.
5. Cryptocurrency payments (Only IDR for now). Payment processing is strictly confined to Indonesian Rupiah (IDR) fiat channels; no blockchain wallets or token payments.
6. AI-based Ticket Price Prediction. Pricing remains entirely static or manually configured by the organizer; no dynamic, algorithmic machine-learning adjustments.
7. Direct Buyer-Seller Communication: The platform does not provide chat, messaging, direct contact information, or communication channels between ticket buyers and sellers. All resale transactions must be completed through platform-managed workflows.
8. Ticket Buyback Guarantee: The platform does not guarantee the sale of resale tickets and is not responsible for purchasing tickets that remain unsold in the resale marketplace.

2.3 Deliverables

List all tangible outputs of this project.

1. Functional website and user dashboard
2. Working integrated payment methods
3. Working admin dashboard
4. QR code ticket generator & validation system
5. Captcha-based bot prevention system

2.4 Constraints

List constraints such as time, budget, technology, or regulatory requirements.

1. The website must handle high amount of traffic (>1000 User)
2. Time must be completed within 1 month
3. The website must be fully stabilized, load-tested, and audited prior to the first major public on-sale date
4. The core codebase must be engineered defensively to accommodate this upcoming asynchronous traffic-throttling architecture without requiring a fundamental rewrite of the checkout logic
5. The technical solution must utilize highly efficient caching layers and strict computing to handle massive traffic spikes cheaply, avoiding over-provisioning expensive, continuously running database servers.
6. The maintenance scheduled activities system shall only be performed outside active event periods and peak ticket sales windows.

3. Stakeholder Identification

List everyone who has interest in or is affected by the project. Use the table below to map stakeholders. You may use a Stakeholder Matrix to visualize influence vs. interest for prioritization.

Stakeholder Name	Role / Title	Interest Level	Influence Level	Communication
Mam Farah Yulianti	Auditor	Highest	High	
Users	Customer / Ticket Buyer	High	Consulted	Email, Notifications, User Dashboard
Event Organizers	Event Creator & Manager	High	Consulted	Organizer Dashboard, Email, Reports
Vendors / Venue Managers	Venue & Event Operations Manager	High	Consulted	Dashboard, Email, Operational Reports
Event Auditors (Admin)	Event Verification Officer	High	Decision Maker	Admin Dashboard, Internal Communication
Super Admin	Platform Administrator	High	Decision Maker	System Dashboard, Internal Communication

4. Business Requirements

Describe the high-level business needs that drive the project. Business requirements answer "What does the business need?" not "How will it be built?" Use the BR-XXX format for each requirement.

Req ID	Description	Priority	Status
BR-001	The system shall provide a centralized platform for event ticket sales and management.	High	Approved
BR-002	The system shall simplify the ticket purchasing process to improve user experience and reduce barriers for new users.	High	
BR-003	The system shall ensure fair ticket distribution among users.	High	
BR-004	The system shall prevent activities that negatively impact ticket availability, including automated purchases, ticket scalping, and fraudulent transactions.	High	
BR-005	The system shall provide a secure environment for user accounts, ticket ownership, and financial transactions.	High	
BR-006	The system shall establish trust by ensuring that all published events have passed a verification and approval process.	High	
BR-007	The system shall support venue management, seating arrangements, and capacity limitations for both seated and standing events.	High	
BR-008	The system shall maintain reliable service during periods of high user traffic and peak ticket sales.	High	
BR-009	The system shall support secure digital ticket ownership and validation mechanisms.	High	

BR-010	The system shall facilitate collaboration among administrators, auditors, vendors, and event organizers through a centralized management environment.	Medium	
BR-011	The system shall reduce operational inefficiencies caused by duplicated information and fragmented workflows.	Medium	
BR-012	The system shall provide a secure ticket resale mechanism that allows users to transfer ticket ownership through the platform.	High	
BR-013	The system shall provide secure and reliable payment processing to support user transactions.	High	
BR-014	The system shall maintain the security and availability of data to support continuous platform operations.	Medium	
BR-015	The system shall support a sustainable revenue model for platform operators and event organizers.	High	
BR-016	The system shall provide easy and convenient access to the system.	High	
5. Functional Requirements			
BR-017	The system shall support ticket purchase limitation policies to ensure fair ticket allocation and reduce scalping activities.	High	
<i>Specify the detailed system and support the purchase must have. Each functional requirement should be testable: you should use a verb to start each requirement. Use the FR-XXX format. Aim for at least 10 functional requirements.</i>			
BR-018	The system shall ensure all ticket resale transactions are conducted within the platform.	High	
Req ID	Description		Priorit
			Status
BR-019 FR-001	The system shall provide a secure and controlled ticket resale marketplace for verified users.	High	High
BR-020 FR-002	The system shall ensure resale ticket prices do not exceed the event organizer's original official ticket price, applied independently of any resale fee, tax, or payment gateway fee charged at checkout.	High	High
BR-021	The system shall allow users to register tickets and bank identities using KTP or Passport information, which will be verified by event staff during check-in.	High	High
BR-022	The system shall act as a ticket resale marketplace and shall not be responsible for purchasing unsold resale tickets.	High	High

5. Functional Requirements

Specify the detailed system requirements for each functional requirement should be testable; you should use a clear and concise language. Use the FR-XXX format. Aim for at least 10 functional requirements.

FR-005	The system shall implement CAPTCHA verification mechanisms during user registration, login attempts after multiple failed authentications, and before virtual queue entry during high-demand ticket sales periods to prevent automated bot activities and unauthorized access.	High	
FR-006	The system shall allow Event Organizers to create, manage, and submit events together with supporting authorization documents.	High	
FR-007	The system shall allow Event Auditors to review, approve, or reject submitted events before publication.	High	
FR-008	The system shall provide venue management features including seating maps, seating arrangements, and venue capacity controls.	High	
FR-009	The system shall allow users to select seats or tickets based on venue configuration and available capacity.	High	
FR-010	The system shall allow Vendors or Event Organizers to define and modify ticket purchase limits per user (e.g., maximum 7 tickets per account).	High	
FR-011	The system shall process payments through an integrated payment gateway (Midtrans) supporting bank transfer/virtual account, QRIS, GoPay, ShopeePay, and credit/debit card, applying method-specific transaction fees, and shall provide payment confirmation upon success..	High	
FR-012	The system shall generate QR-code tickets and unique ticket URLs after successful payment.	High	
FR-013	The system shall provide a virtual queue mechanism during high-demand ticket sales periods to manage user access, maintain system stability, and prevent server overload.	High	
FR-014	The system shall provide an official resale marketplace for users who wish to sell previously purchased tickets.	High	
FR-015	The system shall require all resale transactions to be completed within the platform.	High	
FR-016	The system shall not provide direct communication features between buyers and sellers within the resale marketplace.	High	

FR-017	The system shall automatically validate resale listings to ensure the listed price does not exceed the event organizer's original official ticket price.	High	
FR-018	The system shall automatically verify resale eligibility before publishing a listing, including confirming the ticket is an original (non-resold) ticket. Tickets that have already been resold once shall be rejected from listing.	High	
FR-019	The system shall automatically transfer ticket ownership to the buyer after a successful resale transaction.	High	
FR-020	The system shall distribute resale proceeds only to verified bank accounts registered by users.	High	
FR-021	The system shall allow users to register and verify a bank account for receiving resale proceeds.	High	
FR-022	The system shall automatically calculate and deduct a platform service fee — 4% for primary ticket sales and 2% for resale transactions — inclusive of applicable PPN (11%), from the gross transaction value before transferring net proceeds to the seller's verified bank account. The buyer's payment gateway fee (per selected payment method) shall be charged separately to the buyer at checkout.	High	
FR-023	The system shall calculate PPN (currently 11%) on the platform service fee and payment gateway fee, applicable separately from the ticket face value, in accordance with Indonesian tax regulation.	High	
FR-024	The system shall display an itemized fee breakdown (ticket price, platform fee, payment gateway fee, applicable PPN, total amount due) to the buyer before payment confirmation	High	
FR-025	The system shall allow administrators to configure platform service fee rates without requiring a code deployment (hardcoded fee rates)	Medium	
FR-026	Upon successful completion of a resale transaction, the system shall immediately invalidate the original ticket's QR code and ticket URL issued to the seller, and generate a new QR code and ticket URL bound to the buyer's verified identity.	High	

FR-027	The system shall display an itemized breakdown of resale price, platform service fee, applicable PPN, and payment gateway fee to the buyer prior to resale checkout confirmation	Medium	
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6. Non-Functional Requirements

Define quality attributes: how well the system should perform. Non-functional requirements are often overlooked but cause the most costly issues when ignored. Use the NFR-XXX format. Aim for at least 5 non-functional requirements.

6.1 Performance

Req ID	Description	Priority	Status
NFR-001	The system shall load pages within 3 seconds for at least 95% of requests.	High	
NFR-002	The system shall support at least 1,000 concurrent users without significant performance degradation.	High	
NFR-003	The virtual queue system shall maintain service availability during peak ticket sales periods.	High	

6.2 Security

Req ID	Description	Priority	Status
NFR-004	The system shall securely store user credentials using industry-standard password hashing mechanisms.	High	
NFR-005	The system shall encrypt all sensitive data in transit using HTTPS and TLS protocols.	High	
NFR-006	The system shall encrypt stored identity documents, banking information, and ticket ownership records.	High	
NFR-007	The system shall perform automated daily backups of critical system data.	High	

6.3 Availability

Req ID	Description	Priority	Status
NFR-008	The system shall maintain a minimum uptime of 99.5% during operational periods.	High	Draft

6.4 Usability

Req ID	Description	Priority	Status
NFR-009	New users shall be able to complete the ticket purchasing process within five minutes without prior training.	Medium	Approved
NFR-010	The system shall provide a responsive user interface that supports desktop, laptop, tablet, and mobile devices.	High	Approved
NFR-011	The system shall provide a consistent user experience and navigation structure across all supported devices.	Medium	Approved

7. Assumptions and Constraints

Document conditions assumed to be true and any limitations on the project. Assumptions that prove wrong become risks. Review them regularly.

7.1 Assumptions

List conditions that are believed to be true for this project to proceed.

1. User have Internet access and has a stable connection to it
2. Attendees will prefer digital tickets over manual or paper-based ticketing processes, and vice versa.
3. Venue information is maintained and updated regularly
4. Event organizers provide accurate event details
5. Each ticket is associated with a unique attendee
6. The application will primarily be accessed through modern web browsers and mobile devices
7. Users possess valid email addresses or phone numbers
8. Third-party services (payment gateways, email services, SMS services, QR code generation services) are assumed to be available and functioning correctly

7.2 Constraints

List limitations that bound the project scope, schedule, or resources.

1. **Limited Staff Availability**
 - Event organizers may have a limited number of staff available for ticket verification and venue management during an event.
2. **Peak-Hour Crowd Density**
 - Large numbers of attendees may arrive within a short period before an event starts, creating operational bottlenecks at entry points.
3. **Existing Business Processes**
 - The system must accommodate the organization's existing event planning, ticket sales, and attendee check-in workflows.
4. **Venue Infrastructure Variability**
 - Different venues may have different layouts, capacities, entrances, and operational procedures that the system must support.
5. **Limited Training Time**
 - Event staff may have minimal training before using the system, requiring a simple and intuitive interface.
6. **Multiple Stakeholder Involvement**

- Event organizers, venue managers, security personnel, and attendees have different operational requirements and responsibilities.
- 7. **Operational Hours**
 - System maintenance and updates cannot interfere with active events or peak ticket-selling periods.
- 8. **Emergency Situations**
 - The system must continue supporting attendee verification and attendance tracking during emergency situations such as evacuations or venue disruptions.
- 9. **High Transaction Periods**
 - Popular events may experience sudden spikes in ticket purchases immediately after sales are opened.
- 10. **Compliance with Venue Policies**
 - Ticketing and entry management processes must follow the operational rules established by venue owners.
- 11. **Multi-Event Operations**
 - A venue may host multiple events simultaneously, requiring the system to separate attendee and ticket data accurately.
- 12. **Dependency on Human Verification**
 - Certain situations, such as damaged QR codes or attendee disputes, may require manual intervention by event staff.
- 13. **Geographical Distribution**
 - Events may be conducted across multiple venues in different locations, requiring centralized management.
- 14. **Customer Support Availability**
 - Resolution of ticketing issues may depend on the availability of customer service personnel.
- 15. **Industry Standard Practices**
 - The system must align with common practices used in the event management and ticketing industry.

8. Acceptance Criteria

Define measurable conditions that must be met for the project to be considered complete and accepted by stakeholders. Without clear acceptance criteria, there is no objective way to determine if the project is done.

No.	Acceptance Criterion	Met? (Y/N)	Verified By
1	All functional requirements pass testing with zero critical bugs		
2	System handles the expected number of concurrent users without degradation		
3	User documentation covers all core features		
4	Stakeholder sign-off obtained after demonstration		
5	Venue management features accurately create, update, delete, and display venue information with 100% data consistency during testing.		
6	The system shall verify QR-code-based tickets with an accuracy rate of 100% for valid tickets and reject 100% of duplicate or invalid tickets during testing.		
7	Security requirements are satisfied, including password hashing, role-based access control, input validation, data encryption and HTTPS communication.		
8	All project deliverables, including source code, documentation, database schema, and deployment guide, are completed and accepted.		
9	The system shall send booking confirmation notifications (email or in-app) within 1 minute after a successful ticket purchase.		
10	The system must function correctly on desktop, tablet, and mobile devices without critical UI issues.		
11	System must enforce role-based access control, ensuring that users can only access features permitted by their assigned role.		
12	System shall display real-time ticket availability and update remaining ticket counts immediately after each purchase.		

Appendix: Writing Good Requirements

Requirements should be SMART: Specific, Measurable, Achievable, Relevant, and Time-bound. Below are examples of how to improve vague requirements into well-written ones.

Bad Requirement	Good Requirement	Why It's Better
The system should be fast	The system shall load any page in under 3 seconds on a standard broadband connection	Measurable
Users can log in	The system shall authenticate users via email and password with bcrypt hashing	Specific
The app should look nice	The UI shall follow the approved style guide with consistent spacing and typography	Testable
It should handle lots of users	The system shall support 50 concurrent users without performance degradation	Measurable
Data should be safe	All user data shall be encrypted at rest using AES-256 and in transit using TLS 1.2+	Specific

Sign-Off

This section confirms that the stakeholders have reviewed and approved the contents of this BRD.

Name	Role	Signature	Date